Short-Term PM2.5 Forecasting in Yerevan: A Comparative Study of Statistical and Machine Learning Models

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1. Introduction

Air pollution remains a significant environmental and public health concern in Yerevan, particularly during periods of stagnant weather and increased heating demand. Fine particulate matter (PM2.5) is of special concern due to its well-documented impact on respiratory and cardiovascular health.
Accurate short-term forecasting of PM2.5 levels can support early warnings, improve public awareness, and assist policymakers in understanding pollution dynamics. While forecasting approaches have been widely studied internationally, there is limited systematic comparison of predictive models tailored specifically to Yerevan's environmental conditions.
This project aims to develop and compare a structured set of statistical and machine learning models for short-term PM2.5 forecasting in Yerevan. Rather than focusing on a single complex architecture, the study emphasizes methodological comparison, interpretability, and robustness within realistic computational constraints.

2. Research question

Primary Research Question:
Which forecasting approaches provide the most reliable short-term air pollution (PM2.5) predictions for Yerevan?

Secondary Questions:
Do machine learning and deep learning models significantly outperform classical time-series models?
How sensitive are forecasts to preprocessing choices such as outlier handling and lag selection?

3. Methodology

The research will follow a structured time-series forecasting pipeline, beginning with the collection and consolidation of hourly PM2.5 measurements from publicly available environmental monitoring platforms in Yerevan. Where available, additional explanatory variables such as PM10, NO_2, CO, SO_2, O_3, and meteorological indicators (e.g., temperature, wind speed, humidity) will be incorporated to improve model robustness; however, the availability and continuity of these variables may be limited and will depend on data quality and station coverage. During preprocessing, the dataset will undergo temporal alignment, handling of missing observations through interpolation where appropriate, basic outlier detection using rolling statistical methods, and the construction of lag-based features to capture short-term temporal dependencies. The study will implement a set of baseline, statistical, and machine learning forecasting approaches, including persistence benchmarks, regression-based models, and selected tree-based ensemble methods, with model complexity adjusted according to data characteristics and feasibility within the project timeline. Forecasts will be generated for short-

term horizons (e.g., 1-4 hours ahead) using time-based train-validation-test splits to preserve chronological structure. Model performance will be evaluated using standard regression metrics such as MAE, RMSE, and R^2, with emphasis placed on comparative analysis and interpretability rather than architectural complexity.

4. Expected results

The anticipated outcome of this study is the establishment of a structured performance benchmark for short-term PM2.5 forecasting in Yerevan, providing a comparative assessment of baseline, statistical, and machine learning approaches under local data constraints. The analysis is expected to quantify differences in predictive accuracy across model families and forecasting horizons, identify whether more complex methods yield meaningful improvements over simpler regression-based techniques, and highlight the influence of data availability and preprocessing choices on model performance. The findings will be presented through performance metrics, error analysis, and forecast-versus-actual visualizations, ultimately contributing a reproducible and methodologically transparent framework for air quality forecasting research in Armenia.

5. Timeline and deliverables

February 25 – Literature review completion and dataset consolidation
March 11 – Data preprocessing and feature engineering
March 22 – Implementation of baseline and regression models
April 5 – Implementation of machine learning models
April 19 – Model evaluation and comparative analysis
May 5 – Draft report writing
May 12 – Final report revision and documentation
May 14 – Poster preparation and submission

6. Risks

Although the project follows a structured analytical framework, several potential risks have been identified. Data availability and continuity may pose challenges, particularly if monitoring stations contain gaps, inconsistent measurements, or limited historical coverage. In particular, obtaining complete and reliable data for additional air pollutants (such as NO_2, CO, SO_2, and O_3) as well as meteorological variables (including wind speed, wind direction, and humidity) may prove difficult due to missing observations or fragmented station reporting, which could restrict the scope of multivariate modeling. Model overfitting is another potential risk, especially when working with limited time-series data, and will be mitigated through time-based validation and appropriate regularization techniques. Finally, given the project timeline, computational constraints and implementation complexity may limit the number of models explored; therefore, the study prioritizes methodological rigor, reproducibility, and comparative clarity over excessive architectural expansion.

7. References

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